

CareerGPS

Requirements Specification Document

Your AI-Powered Career Navigation Platform

Document Type	Requirements Specification
Version	1.0 — Initial Release
Program	Rwaad Masr Digital Scholarship
Team Size	12 Members (4 Frontend 2 Full-Stack 6 Agentic AI)
Date	April 2025

1. Introduction

1.1 Project Overview

CareerGPS is an AI-powered web platform that transforms career development by generating personalized learning roadmaps based on each user's CV and existing skills. Unlike generic learning platforms, CareerGPS analyzes the user's real background and recommends a tailored step-by-step path — with a specific course for every skill, so learners start immediately without getting lost in the noise. A built-in learning calendar with motivational push notifications keeps users accountable, while gamification elements like XP, badges, and levels make the journey feel engaging rather than overwhelming. Advanced features include AI-powered skill assessments, mock interview simulations, an ATS-friendly CV builder, and a first-of-its-kind **Career Simulation** mode that lets users experience a career hands-on before committing to it. CareerGPS doesn't just point you in the right direction — it walks with you every step of the way.

1.2 Problem Statement

Most online learners struggle with three core problems:

- Not knowing where to start — generic roadmaps ignore individual backgrounds.
- Getting lost in resources — thousands of courses with no guidance on what fits their level.
- Lack of accountability — no structured schedule or motivation to maintain consistency.

1.3 Document Purpose & Scope

This document defines the complete requirements for CareerGPS version 1.0 including stakeholder analysis, user stories, functional requirements, and non-functional requirements. It serves as the primary reference for the development team, project supervisor, and evaluators.

1.4 Team Roles

Role	Members	Responsibilities
Frontend Developers	4 members	UI/UX implementation, React components, responsive design, accessibility
Full-Stack Developers	2 members	API development, backend logic, database integration, authentication
Agentic AI Team	6 members	AI pipeline design, roadmap generation, CV parsing, interview simulation, chatbot, Check CV is ATS friendly and give ascore

2. Stakeholder Analysis

The following stakeholders have been identified as having a direct or indirect interest in the CareerGPS platform:

Stakeholder	Type	Primary Interest	Influence	Priority
Job Seekers (Students & Graduates)	Primary User	Get a personalized, actionable career roadmap based on their actual skills and background.	High	P1
Career Changers	Primary User	Explore and transition into new career paths with guided simulation and learning plans.	High	P1
Course Providers (Coursera, Udemy, etc.)	External Partner	Drive enrollment traffic to their courses via recommendations.	Medium	P2
Recruitment Companies	External Partner	Access a pipeline of job-ready candidates for open positions.	Medium	P2

3. User Stories

User stories follow the standard format: "As a [actor], I want to [action] so that [benefit]." Stories are organized by development phase using MoSCoW prioritization.

Phase 1 — MVP (Must Have)

ID	Actor	User Story	Priority	Phase
US-05	Job Seeker	I want to select a target career path so the system generates a roadmap tailored to my background.	Must Have	Phase 1
US-06	Job Seeker	I want to receive a personalized step-by-step roadmap so I know exactly what to learn next.	Must Have	Phase 1
US-07	Job Seeker	I want to see a course recommendation for each skill in my roadmap so I can start learning immediately.	Must Have	Phase 1
US-08	Job Seeker	I want to set learning days in a calendar so I can plan my study schedule.	Must Have	Phase 1
US-09	Job Seeker	I want to receive motivational push notifications on my study days so I stay on track.	Must Have	Phase 1
US-10	Job Seeker	I want to mark a skill as completed so the system updates my roadmap progress.	Must Have	Phase 1

Phase 2 — Intelligence (Should Have)

ID	Actor	User Story	Priority	Phase
US-11	Job Seeker	I want to take a skill exam so the system can assess my actual knowledge level per skill.	Should Have	Phase 2
US-12	Job Seeker	I want to receive gap analysis results with recommended resources so I can address weaknesses.	Should Have	Phase 2
US-13	Job Seeker	I want to upload a job description and generate an ATS-friendly CV so I increase my interview chances.	Should Have	Phase 2
US-14	Job Seeker	I want to use an AI chatbot to ask questions about my roadmap or the platform so I get instant help.	Should Have	Phase 2
US-15	Job Seeker	I want to compare multiple YouTube playlists so I can choose the best learning resource for each skill.	Should Have	Phase 2
US-24	Admin	I want to update course recommendations and roadmap templates so the content stays current.	Should Have	Phase 2

Phase 3 — Engagement

ID	Actor	User Story	Priority	Phase
US-16	Job Seeker	I want to earn XP points and badges when I complete skills so I feel motivated to keep learning.	Should Have	Phase 3
US-17	Job Seeker	I want to see a leaderboard so I can compare my progress with other learners.	Could Have	Phase 3
US-18	Job Seeker	I want to simulate a job interview with AI so I can practice before real interviews.	Should Have	Phase 3
US-19	Job Seeker	I want the AI to generate a LinkedIn post about my GitHub project so I can showcase my work easily.	Could Have	Phase 3

Phase 4 — Discovery & Jobs

ID	Actor	User Story	Priority	Phase
US-20	Career Explorer	I want to try a hands-on career simulation before choosing my path so I make an informed decision.	Should Have	Phase 4
US-21	Job Seeker	I want to see job listings matching my current skills from recruitment platforms so I know when I'm ready.	Could Have	Phase 4
US-22	Freelancer	I want to see freelance projects matching my skills so I can earn money while learning.	Won't Have (v1)	Phase 5

4. Functional Requirements

Functional requirements define what the system must do. Each requirement is assigned a unique ID, mapped to a system module, and prioritized using MoSCoW.

ID	Module	Requirement Description	Priority
FR-01	Profile	System shall parse uploaded PDF CVs and extract skills, experience, and education using AI.	Must Have
FR-02	Profile	System shall integrate with LinkedIn public data via user-provided URL or exported PDF.	Must Have
FR-03	Profile	System shall integrate with GitHub API to fetch public repositories and infer technical skills.	Must Have
FR-04	Profile	System shall allow users to manually add, edit, or remove skills with proficiency levels.	Must Have
FR-05	Roadmap	System shall generate a personalized, ordered roadmap based on user skills, experience level, and selected career path.	Must Have
FR-06	Roadmap	System shall display roadmap progress with completed, in-progress, and upcoming skills.	Must Have
FR-07	Roadmap	System shall regenerate roadmap if user changes career path or updates skills.	Must Have
FR-08	Courses	System shall recommend at least one course per skill node in the roadmap from curated sources.	Must Have
FR-09	Courses	System shall allow users to mark a course as completed and update roadmap status accordingly.	Must Have
FR-10	Calendar	System shall provide a weekly learning calendar where users can set study days and hours.	Must Have
FR-11	Notifications	System shall send push notifications on scheduled study days with motivational messages.	Must Have
FR-12	Assessments	System shall generate skill-based exams (MCQ + open-ended) for any skill in the roadmap.	Should Have
FR-13	Assessments	System shall analyze exam results, identify skill gaps, and recommend targeted resources.	Should Have
FR-14	CV Builder	System shall accept a job description and generate a tailored, ATS-optimized CV with relevant keywords.	Should Have
FR-15	CV Builder	System shall generate a new CV from scratch for users who do not have one.	Should Have
FR-16	Interview	System shall simulate a job interview by asking role-specific questions and providing AI feedback on answers.	Should Have
FR-17	Chatbot	System shall provide a conversational AI assistant that answers questions about the roadmap and platform features.	Should Have
FR-18	Gamification	System shall award XP points and achievement badges when users complete skills, exams, or milestones.	Should Have

ID	Module	Requirement Description	Priority
FR-19	Gamification	System shall display a leaderboard ranking users by XP within the platform community.	Could Have
FR-20	Career Sim	System shall provide interactive, hands-on career simulations allowing users to experience a career before choosing it.	Should Have
FR-21	LinkedIn AI	System shall generate a LinkedIn post draft about a selected GitHub project for user review before publishing.	Could Have
FR-22	Playlist	System shall compare two or more YouTube playlists and recommend the best match for the user's skill level.	Could Have
FR-23	Jobs	System shall match and display relevant job listings from recruitment platforms based on the user's current skill set.	Could Have
FR-24	Auth	System shall support user registration, login, and profile management with secure authentication.	Must Have

5. Non-Functional Requirements

Non-functional requirements define the quality attributes and constraints the system must satisfy.

ID	Category	Requirement Description
NFR-01	Performance	Roadmap generation response time shall not exceed 5 min under normal load.
NFR-02	Performance	CV parsing and skill extraction shall complete within 2 min for files up to 5MB.
NFR-03	Scalability	System shall support up to 10,000 concurrent users without performance degradation.
NFR-04	Availability	Platform uptime shall be 99.5% or higher, measured monthly.
NFR-05	Security	All user data shall be encrypted at rest (AES-256) and in transit (TLS 1.3).
NFR-06	Security	System shall implement JWT-based authentication with refresh token rotation.
NFR-07	Security	System shall comply with GDPR data privacy requirements, including user data deletion rights.
NFR-08	Usability	Core user tasks (upload CV, generate roadmap) shall be achievable in 3 steps or fewer.
NFR-09	Usability	Platform shall be fully responsive and functional on mobile, tablet, and desktop devices.
NFR-10	Usability	Platform shall support both English and Arabic (RTL) languages.
NFR-11	Accessibility	UI shall meet WCAG 2.1 Level AA accessibility standards.
NFR-12	Reliability	AI-generated roadmaps shall produce consistent, deterministic results for the same user profile and career input.
NFR-13	Maintainability	System shall follow a modular microservices architecture to allow independent deployment of features.
NFR-14	Storage	System shall store user uploaded files (CVs, images) with automatic virus scanning before processing.
NFR-15	API	All public APIs shall follow RESTful conventions with versioning (e.g., /api/v1/) and return standardized JSON responses.

6. Assumptions & Constraints

6.1 Assumptions

- Users have internet access and a modern web browser.
- CVs are in PDF format (English or Arabic).
- LinkedIn and GitHub accounts are public or users provide exported data.
- AI-generated roadmaps require a valid Anthropic API key with sufficient quota.
- Push notifications require users to grant browser/device permissions.

6.2 Technical Constraints

- LinkedIn does not allow automated scraping; integration relies on user-exported PDF or public URL.
- AI roadmap generation depends on third-party API availability (Anthropic Claude API).
- File upload size is limited to 5MB per CV to ensure fast processing.
- The system targets modern browsers (Chrome 90+, Firefox 88+, Safari 14+).

6.3 Business Constraints

- MVP must be completed within 4 months to meet graduation deadline.
- Initial budget covers development tools, hosting, and API costs only.
- Platform will launch in beta with a limited user group before public release.

7. Glossary

Term	Definition
Roadmap	A personalized, ordered sequence of skills and learning tasks tailored to a user's career goal and current background.
Skill Gap	The difference between the skills a user currently has and the skills required for their target career.
ATS	Applicant Tracking System — software used by employers to screen CVs based on keywords before human review.
Career Simulation	An interactive, guided task-based experience that lets a user 'try out' a career path before committing to it.
MoSCoW	Prioritization framework: Must Have, Should Have, Could Have, Won't Have (for this version).
XP (Experience Points)	Gamification points awarded to users when they complete skills, exams, or milestones on the platform.
DFD	Data Flow Diagram — a visual representation of how data moves through the system.
ERD	Entity-Relationship Diagram — a database design diagram showing entities, attributes, and relationships.
JWT	JSON Web Token — a compact, self-contained token used for secure user authentication.

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